

Pranshu Nijhawan

Principal Architect I - Agentic AI & Distributed Systems

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Summary

Principal Architect with 10+ years in enterprise **SaaS**. Currently Enterprise Architect of a **multi-tenant SaaS** platform. Recent work spans AI agent platforms (MCP, **multi-agent orchestration**, **evaluation harnesses**), high-performance data systems (sub-2 second response on 2M+ records, billion-row analytics on **ClickHouse**), and two-layer security architecture (FAC + DAC with **Casbin**). Coached org-wide **Cursor** and AI-assisted engineering practices, delivering a 37% capacity increase, and led **Spec-Driven Development** to evolve the organization toward an **AI-led SDLC**.

- Core: **TypeScript**, Python, C#, .NET, **Node.js**, **React**, **Databricks**, **ClickHouse**, **PostgreSQL**, Kubernetes, Azure, AWS.
- Agentic: **Multi-agent orchestration**, MCP (servers + clients), **Cursor** (Rules / Skills / Agents / Hooks / SDK), SDD (**OpenSpec**, **SpecKit**, **BMad**), **RAG**, evaluation frameworks, T5 / **PyTorch** fine-tuning, **NL-to-SQL**.

Experience

McKinsey & Company

March 2022 - Present

Principal Architect I

Previously: Senior Software Engineer II, Senior Software Engineer I

Enterprise Architect of **Promotion Advisor**, a flagship **multi-tenant SaaS** in McKinsey's **Periscope** suite. Built the platform's multi-agent AI architecture, coached org-wide AI-assisted engineering that delivered a 37% capacity increase, and spearheaded **Spec-Driven Development** to evolve the organization toward an **AI-led SDLC**.

- **Agentic AI Architecture**: Built a multi-agent architecture for the platform using MCP (**Model Context Protocol**) connectors, externalized instruction sets owned by product teams, **evaluation harnesses**, and read-replica isolation with data-access control enforced at the SQL layer. Cut business-rule changes from multi-day deploys to hour-level config updates.
- **Spec-Driven Development & AI-Led SDLC**: Introduced org-wide adoption of **Cursor** and **Spec-Driven Development** (**OpenSpec**, **SpecKit**) across greenfield and brownfield product development. Developed the '**Spec-Driven Development Agent**' and '**Repo Mastery Agent**'-autonomous systems enforcing 11-dimension anti-hallucination audits and 100/100 Spec Quality Scores. This established the proper abstraction and technical detail required for Devs, Product Managers, and QAs to seamlessly collaborate and evolve the organization toward an **AI-led SDLC**.
- **Enterprise Agent Skills Library**: Built a comprehensive library of autonomous, self-contained AI agents utilizing embedded design systems, deterministic validation checklists, and MCP tools. Deployed 10+ production agents across multiple enterprise products, including Architecture Diagram Generators, ADR/APR Writers, and automated Technical Communications.
- **NL-to-SQL ML Prototyping**: Trained a custom encoder-decoder transformer (BPE tokenizer, **PyTorch**) and fine-tuned T5 on synthetic **NL-to-SQL** datasets generated from production **PostgreSQL** schemas.
- **Real-Time Conflict Engine (5,000x Speedup)**: Built real-time multi-dimensional conflict detection on **PostgreSQL** using progressive candidate narrowing. Delivered sub-2 second response times on 2M+ records, achieving a 5,000x algorithmic improvement over the prior implementation.
- **ClickHouse Analytics Platform**: Built the platform's **ClickHouse** analytics layer for billion-row sub-second queries. Optimized the data model with **Projections** and **Dictionaries**, and built a dynamic Query Builder Engine. Delivered 6.6-12.6x cost efficiency vs **Databricks** SQL Serverless and ~11x compression vs Delta Lake.
- **Analytics Cost Migration (\$172K/Year Savings)**: Migrated heavy analytics workloads from variable-cost **Databricks** compute to fixed-cost **PostgreSQL**. Cut annual analytics infrastructure spend by \$172K (a 23-50% reduction depending on the workload).
- **Two-Layer Authorization Engine (FAC + DAC)**: Built a two-layer authorization engine. Layer 1 (**Feature Access Control**) enforces at the **GraphQL** resolver middleware. Layer 2 (**Data Access Control**) integrates **Casbin** at the SQL layer, dynamically rewriting queries so unauthorized data never leaves the database.
- **Engineering Standards & FinOps**: Institutionalized cross-product standards by publishing the C4 Architecture Handbook, deploying a serverless Azure cost attribution dashboard via GitHub Actions, and building an automated external API documentation pipeline with Git changelog guardrails.

Technologies: MCP, Cursor, OpenSpec, SpecKit, BMad, PyTorch, T5, TypeScript, Python, Node.js, Fastify, C#, .NET, React, GraphQL, PostgreSQL, ClickHouse, Databricks, Casbin, Kubernetes, Azure, AWS

Eptura (formerly Condeco)

April 2021 - March 2022

Principal Engineer

Re-architected a single-tenant workspace **SaaS** to cloud-native microservices on AKS and built a global IoT release orchestrator synchronizing 300M+ devices. Directed 20 engineers across 4 squads.

- **Cloud-Native Re-Architecture:** Migrated legacy single-tenant **SaaS** to cloud-native microservices on AKS (**Azure Kubernetes Service**): 60% faster incident recovery, eliminated single-tenant scaling bottlenecks.
- **Global IoT Release Orchestration:** Built a release orchestration system (Azure IoT Hub, Azure Functions, Azure CosmosDB) synchronizing firmware updates across 300M+ IoT devices globally with zero-downtime rollouts.
- **Engineering Leadership:** Directed 20 engineers (4 tech leads) across four squads. Established a structured mentorship program: +20% retention, accelerated engineering progression.

Technologies: C#, .NET, React, Angular, AKS, Azure IoT Hub, Azure Functions, Azure CosmosDB, Azure, SQL Server

Nagarro

December 2019 - April 2021

Senior Engineer

Shipped full-stack web and mobile apps for e-commerce and HR enterprise clients on **Angular**, **.NET Core**, **PostgreSQL**, and **Ionic**. CI/CD and IaC practices cut deployment failures by 70% across client projects.

- **Full-Stack Web + Mobile Delivery:** Shipped production apps on **Angular**, **.NET Core**, **PostgreSQL**, and **Ionic Framework** with 99.5%+ uptime across web and mobile.
- **CI/CD & Infrastructure as Code:** Established CI/CD pipelines and IaC practices: 70% reduction in deployment failures across client projects.

Technologies: C#, .NET Core, Angular, Ionic Framework, PostgreSQL, DynamoDB, AWS, Fargate

Eptura (formerly Condeco)

September 2018 - December 2019

Software Engineer

Shipped C# / .NET / **React** / SQL Server microservices that sustained 3x customer growth without reliability regressions. Cut build times by 40% and core query latency by 3x via data model refactoring.

- **Microservices for 3x Customer Scale:** Designed and shipped C# / .NET / **React** / SQL Server microservices, sustaining 3x customer scale-up without reliability regressions.
- **CI/CD & Data Model Optimization:** Cut build times by 40%; refactored core data models to reduce average query latency by 3x.

Technologies: C#, .NET, React, Angular, SQL Server, Azure

Gartner

February 2016 - August 2018

Associate Software Engineer

Previously: Intern

Shipped backend features on C# / .NET Core / **React** for an HR analytics **SaaS**, improving survey data accuracy by 60% for consulting clients. Automated SQL workflows saved 240+ engineering hours annually.

- **SaaS Analytics Backend:** Shipped backend features on C# / .NET Core / **React**: improved survey data accuracy by 60% for HR consulting clients.
- **SQL Workflow Automation:** Automated SQL workflows (stored procedures, SQL Agent jobs): saved 240+ engineering hours annually.

Technologies: C#, .NET Core, React, SQL Server, AWS

Core Competencies

- **AI & Agentic Engineering:** Agentic AI Architecture, Multi-Agent Orchestration, MCP (Model Context Protocol), RAG (Retrieval-Augmented Generation), Prompt Engineering, Evaluation Frameworks, Cursor (Rules, Skills, Agents, Hooks, SDK), Spec-Driven Development (OpenSpec, SpecKit, BMad), Instruction Sets, Knowledge Base Grounding, Encoder-Decoder Transformers, T5 Fine-tuning, PyTorch, BPE Tokenization, NL-to-SQL.
- **SaaS & Architecture:** System Design, Microservices, Event-Driven Architecture, Domain-Driven Design (DDD), API Design (REST, GraphQL), Serverless, Distributed Systems, C4 Modeling, ADR / APR Authorship, Data Lakehouse, Multi-Tenant SaaS.
- **Cloud & Infrastructure:** Azure, AWS, Kubernetes (AKS), Terraform, ArgoCD, Helm, GitOps, CI/CD, Infrastructure as Code (IaC), Observability (Dynatrace, OpenTelemetry), Security Scanning (Wiz, SonarQube).
- **Data Engineering & Real-Time Analytics:** Databricks, Spark (PySpark, SparkSQL), ClickHouse, PostgreSQL, Real-time Analytics, ELT / ETL, Blue-Green Deployments, Data Modeling, Columnar Analytics, FinOps / Cost Optimization.
- **Tech Stack:** TypeScript, Python, C#, .NET Core, Node.js, Fastify, React, Angular, SQL.
- **Security & Compliance:** SOC2, InfoSec, Secure SDLC, Casbin RBAC, Two-Layer FAC + DAC Architecture, Centralized Identity Integration, Audit Automation, SLA / SLO Management.
- **Leadership & Mentorship:** Engineering Mentorship, Architecture Ownership, Cross-functional Stakeholder Management, Technical Coaching (Cursor / Agentic Engineering / SDD), Tech Lead, Cross-Product Influence.

Certifications

- [Microsoft Certified: Azure Data Engineer Associate](#)
- [Microsoft Certified: Azure Developer Associate](#)

Educational Qualifications

Gurgaon Institute of Technology & Management

2012 - 2016

Bachelor of Technology (B. Tech), Computer Science & Engineering